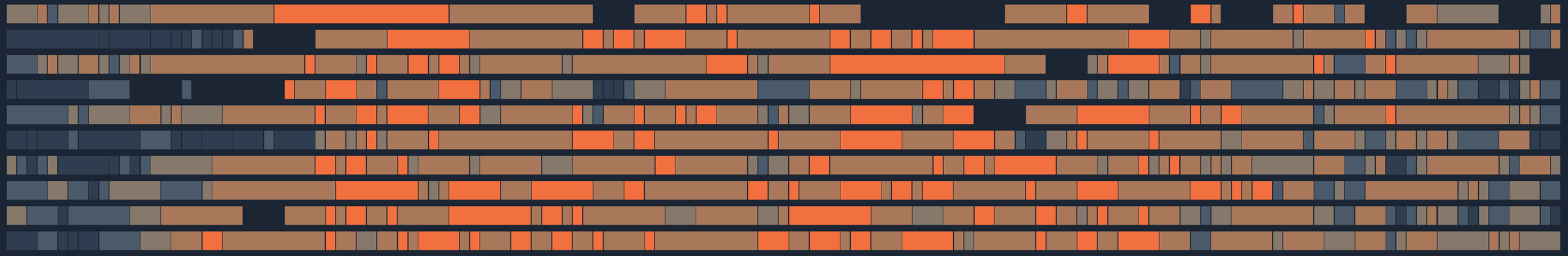
ASM PROJECT 2026M · STOCHASTIC MODELLING · AIR POLLUTION

How often does Delhi's air turn Severe, and how long does it stay that way?

Poisson processes and Markov chains on ten years of CPCB PM2.5 data, R K Puram

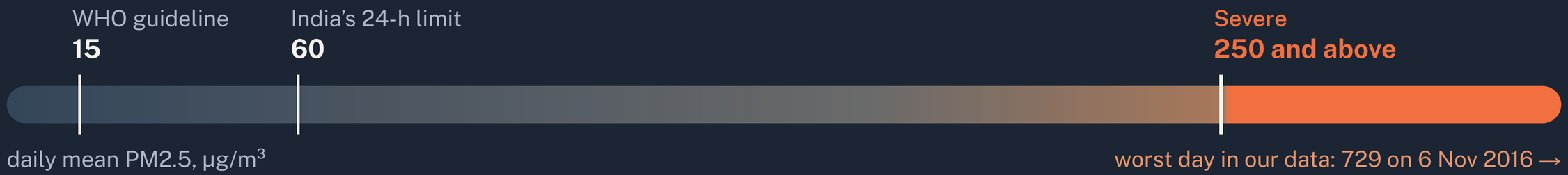


Each row is one winter, 1 Oct → 28 Feb, 2016/17 (top) to 2025/26 · Good–Satisfactory Moderate Poor Very Poor Severe

Harsh Raj · data & Poisson **Vivian Mathew** · Markov chain **Rudraksh Kapil** · simulation & validation

THE PROBLEM

Severe means 17 times the safe limit



17×

the WHO 24-hour guideline at the Severe line (250 ÷ 15)

≈ 32

Severe days a year at R K Puram, 2016–2025

17 days

longest Severe run in a row, from 20 Dec 2018

1 · POISSON

How many Severe days, and when?

2 · MARKOV

Once it turns Severe, how long does it last?

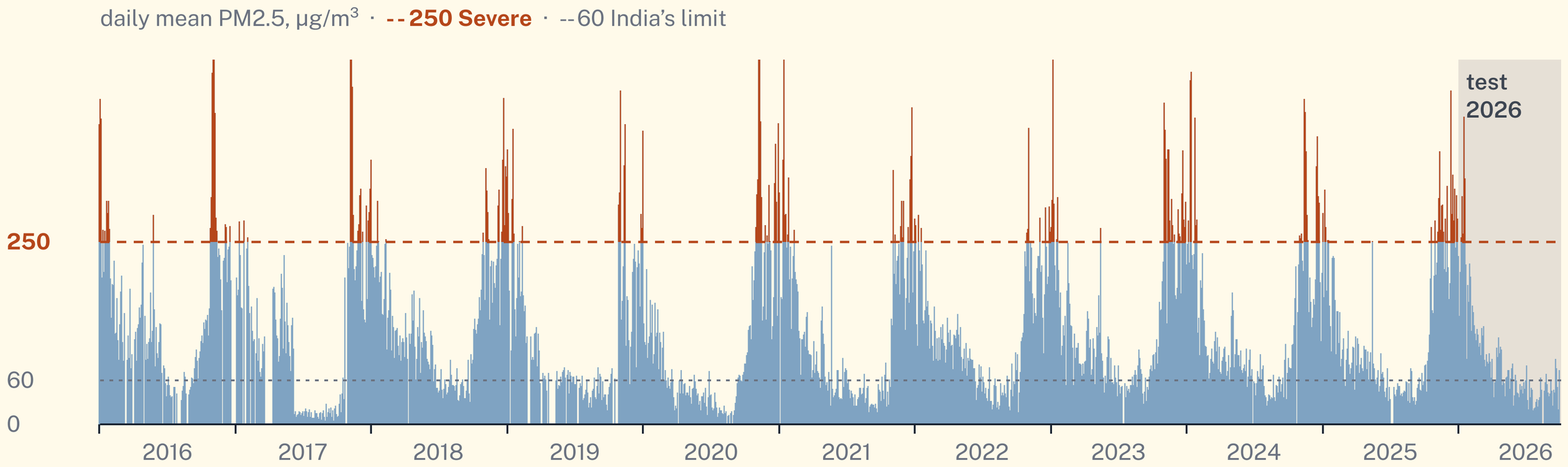
3 · SIMULATION

Can the past predict new years?

WHO 2021 guideline: 15 µg/m³ (24-h); India NAAQS: 60 µg/m³ (24-h)

DATA · CPCB CAAQMS, R K PURAM (DPCC SITE 124)

A spike above 250 every single winter



≈ 80,000 hourly readings → **3,430** valid days → **5** AQI bands **(Markov)** → **300** Severe days **(Poisson)**

A day counts if ≥ 18 of 24 hours are valid · 89% observed · 5% short gaps filled · 6% excluded · 2026 kept for testing

HOMOGENEOUS POISSON PROCESS

About 32 Severe days a year, in clumps

31.9

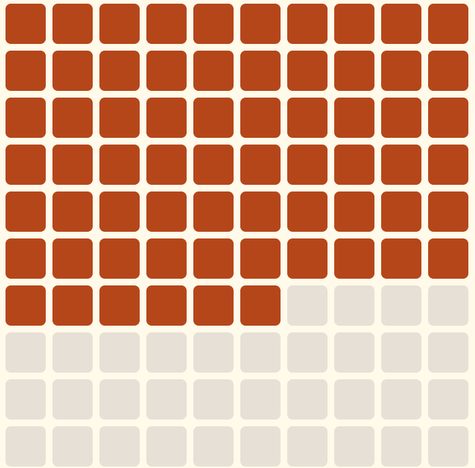
Severe days a year

$\hat{\lambda} = n / T = 300 / 3{,}430$ valid days
exact 95% CI: 28.4 – 35.8 a year

If Severe days were a Poisson process, gaps between them would be Exp(λ) and **one-day gaps would be rare**.

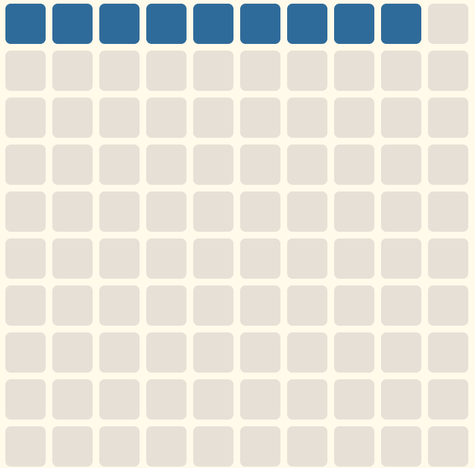
Of every 100 gaps between Severe days, how many are **just one day?**

REAL DATA



66 of 100

IF POISSON WERE RIGHT

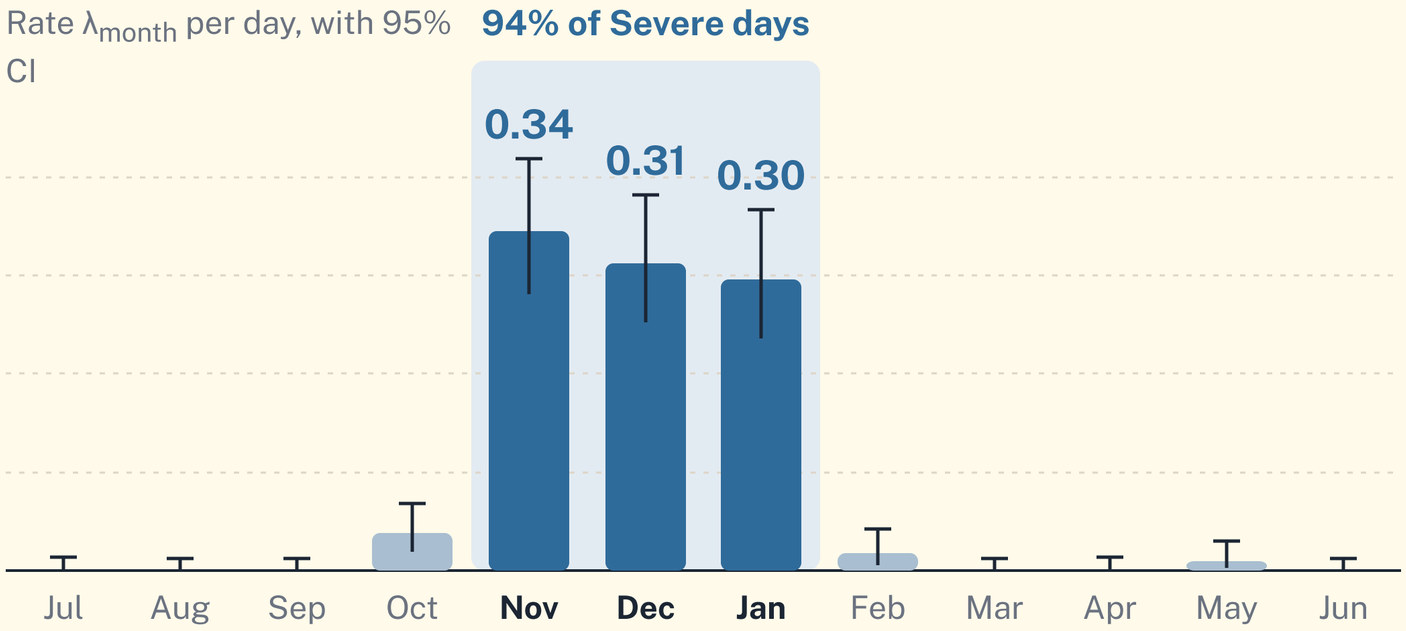


9 of 100

REJECTED Monthly variance is **8.1×** the mean (a Poisson process needs 1×), and Severe days clump together.

SEASONAL POISSON PROCESS

94% of Severe days fall in Nov-Jan



AIC 928 → 270

far better fit than one constant rate: LR = 680 on 11 df

-0.4% a year

no significant trend, 95% CI −5.6% to +5.0%

95.6%

of unseen 2023–26 months fell inside the 95% interval

WHY MEMORY Inside a winter month, Severe days arrive in runs, so tomorrow's air should depend on **today's**.

MARKOV CHAIN

Tomorrow's air depends on today's

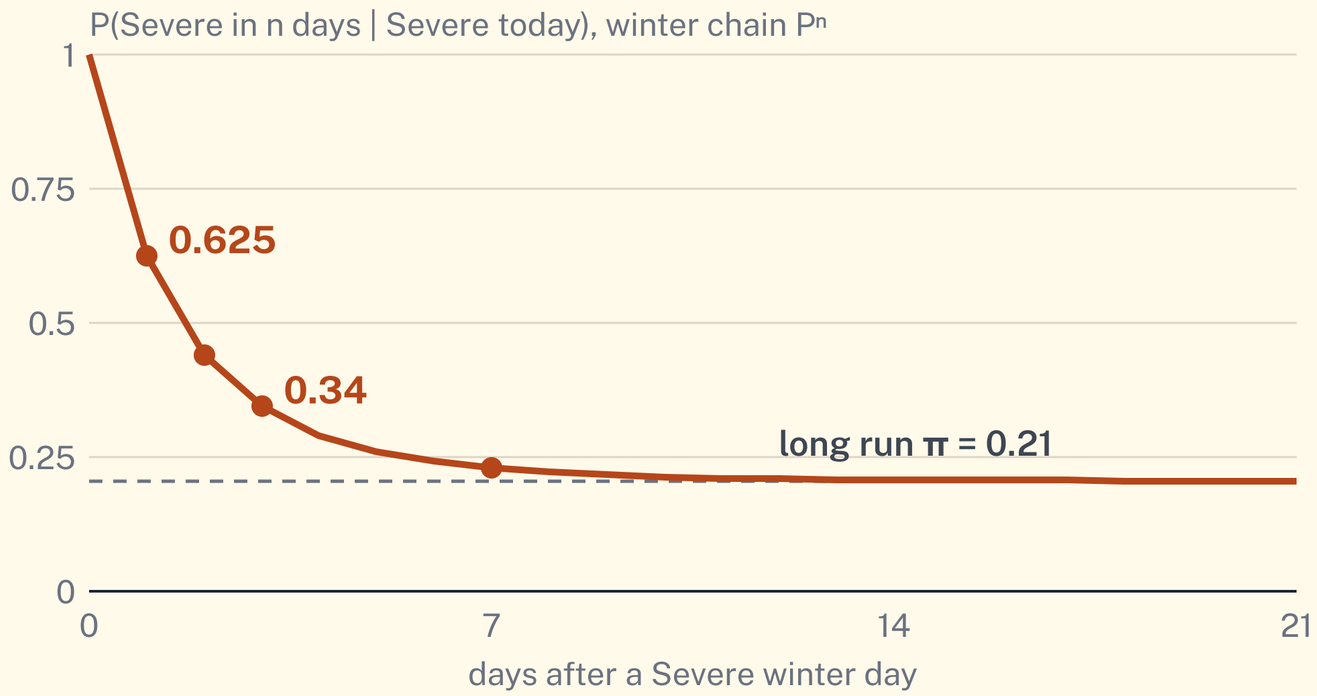
X_t = today’s AQI band, one step = one day. MLE $\hat{p}_{ij} = n_{ij}/n_i$ from 3,394 transitions. Only 3.5% of days jump two or more bands.



All-year matrix, 2016–2025

WHAT THE CHAIN SAYS

Memory of a Severe day fades in a week



2.7 d
mean Severe spell 1/(1 − p_{SS}); data 2.68

1 in 5
winter days Severe in the long run, π = 0.21

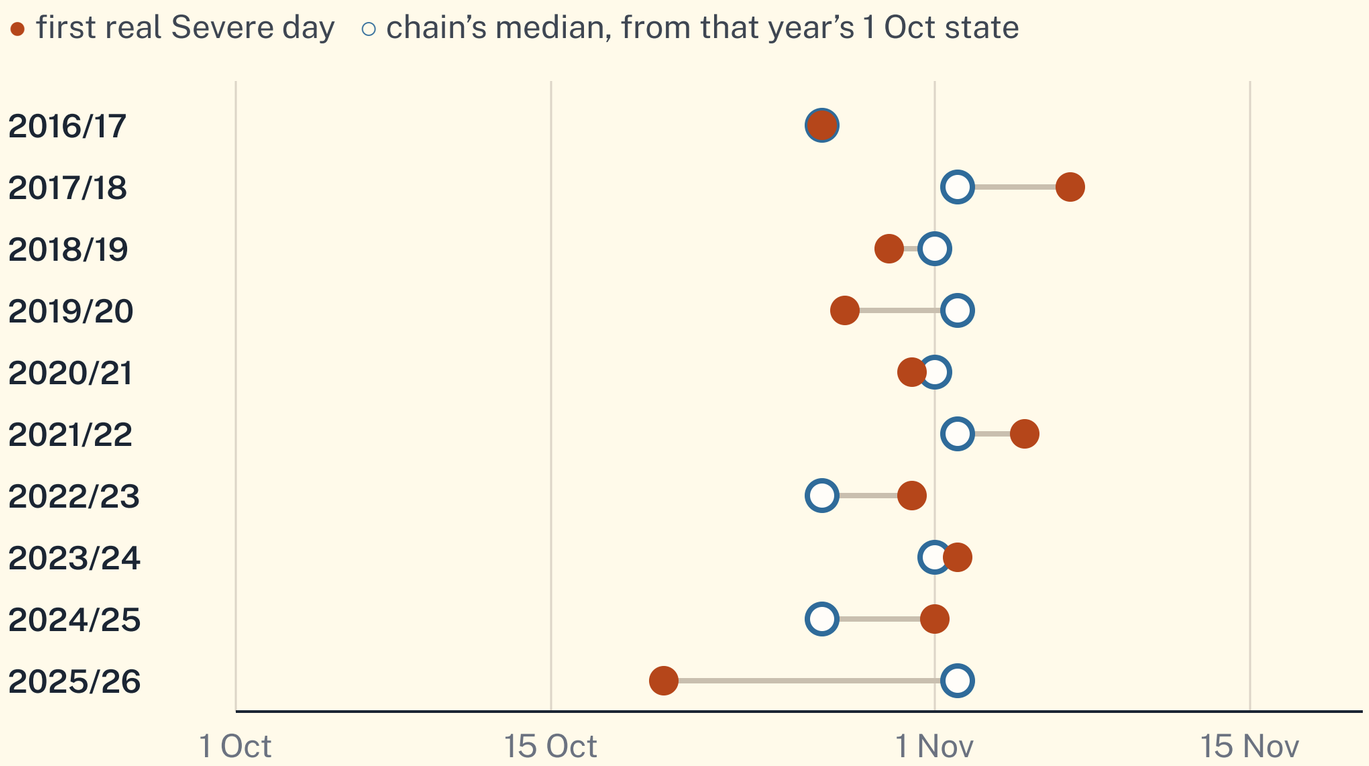
10 d
mean wait for Severe from a Very Poor winter day

p = 0.69
2nd-order chain adds nothing in winter (G^2 = 39.8, 45 df)

Winter and summer matrices differ ($p \approx 10^{-20}$), so the chain is fitted per season.

FIRST-PASSAGE TIMES

The smog season starts around 1 November



30 days

median real wait from 1 Oct to the first Severe day (range 19–37)

10 of 10

seasons had their first Severe day by 15 Nov; the chain gives 87–93%

29 vs 29.6

days: the monthly chain's mean wait from a Moderate 1 Oct (the usual start) vs the real mean

First Severe day after 1 October, seasons 2016/17 – 2025/26

METHODOLOGY AND CODE

One engine simulates and tests both models

```
notebooks/04 · StochasticAirQualitySimulator
# advance 2,000 simulated paths by one day
P   = P_season[season[t]]        # 5×5 matrix
cdf = P[states[:, t-1]].cumsum(axis=1)
u   = rng.random((n_sim, 1))
states[:, t] = (u > cdf).sum(axis=1)

# Poisson: independent daily counts
N = rng.poisson(lam[month] * valid_day)

# h-day forecast (Chapman-Kolmogorov)
p = matrix_power(P, h)[x_today, SEVERE]
```

1 **Check the engine**

Simulate from a known P and re-estimate it: bias → 0, SE ∝ 1/√N, 95% CIs cover ≈ 93–95%

2 **Replay the decade**

2,000 synthetic 2016–25 decades per model, same calendar and missing days, seed 2026

3 **Forecast unseen days**

Refit on 2016–22; score every day of 2023 – Sep 2026 by Brier score and AUC

github.com/HarshRaj797864/asm-air-pollution

SIMULATION · 2,000 SYNTHETIC DECADES PER MODEL

Each model gets part of the decade right

Feature (2016–2025)	Real data	Monthly Poisson	Markov chain
Total Severe days	300	✓ 301 (269–334)	✓ 299 (245–354)
Mean Severe spell	2.68 d	✗ 1.34 d, too short	✓ 2.64 d
Longest Dec–Feb spell, mean	6.3 d	✗ ≈ 2.5 d, too short	✓ 5.4 d
Month-to-month variation	1.77	✗ 1.11, too steady	✗ 6.8, too bursty

Spells: Markov vs real lengths, two-sample KS p = 0.32, so they are indistinguishable

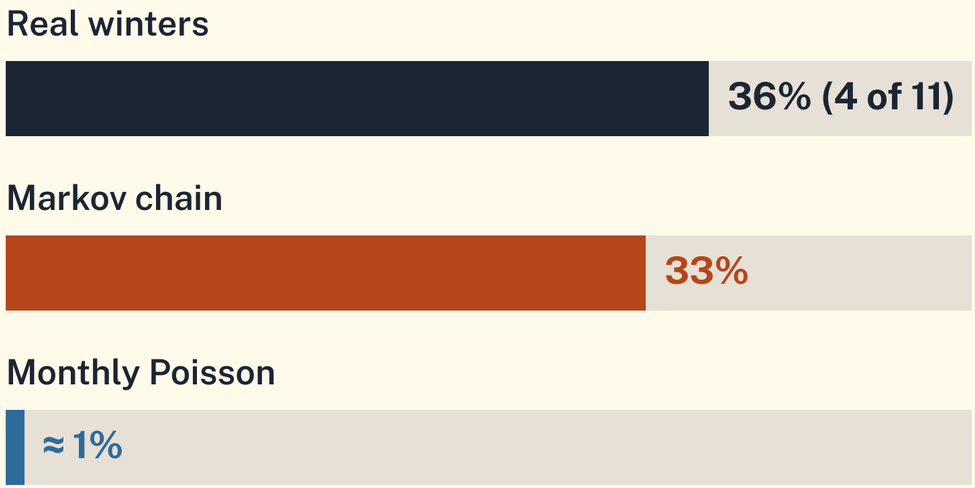
Variation: no model matches 1.77, so the truth sits between “no memory” and “simple memory”

Ranges are 95% of simulations

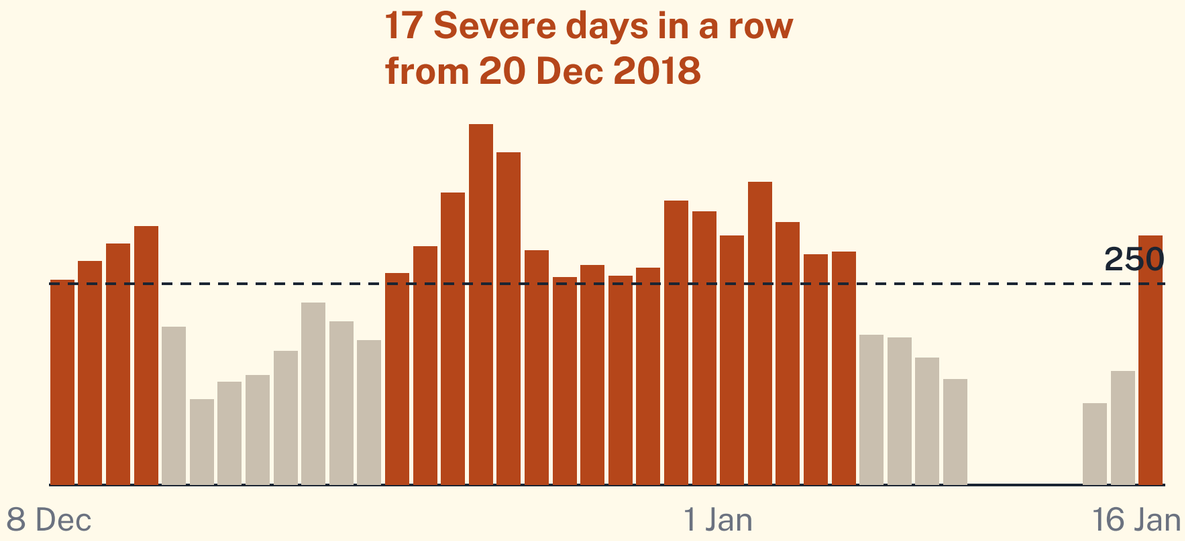
EXTREME SPELLS · SIMULATED DEC–FEB WINTERS

Only memory produces week-long smog

Dec–Feb winters with a Severe run of 7+ days



Daily PM2.5, 8 Dec 2018 – 16 Jan 2019

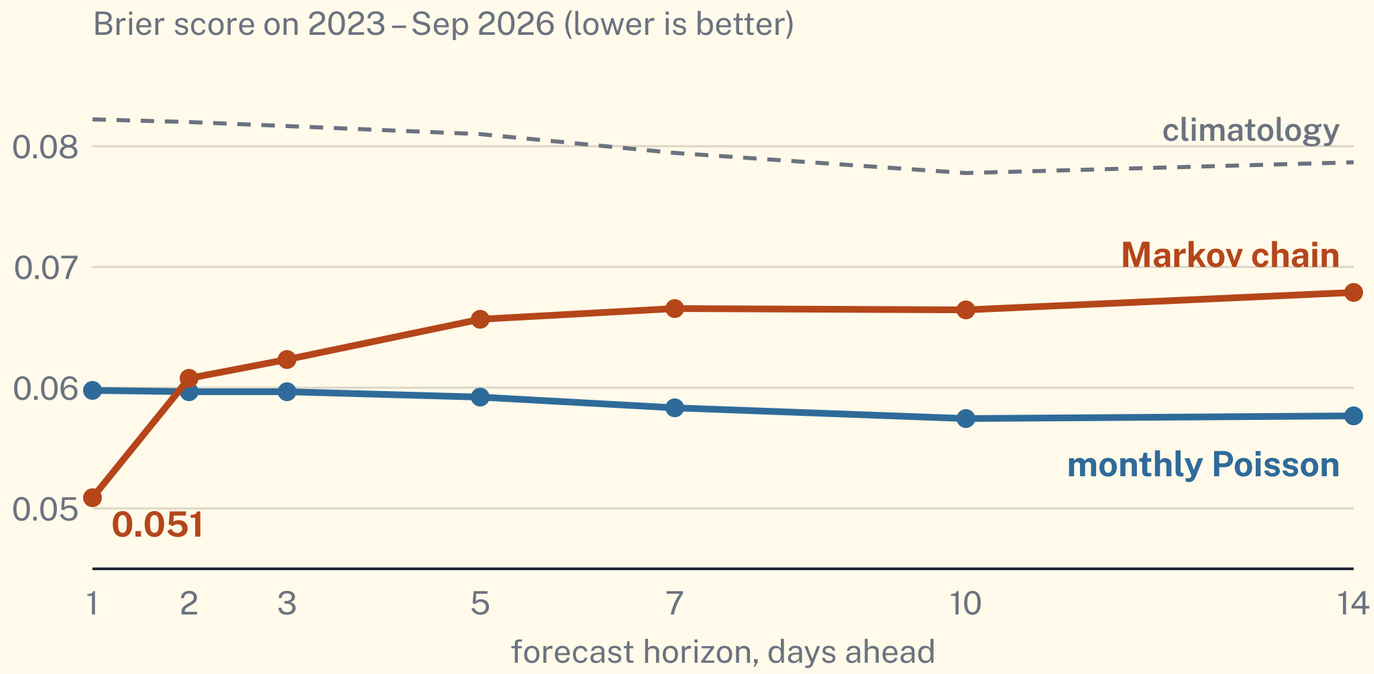


Even the chain makes a 17-day run rare: the real tail is heavier still.

20,000 simulated winters per model

OUT-OF-SAMPLE TEST · FIT 2016–22

Markov wins tomorrow; Poisson after that



1 day ahead: Brier 0.051 vs 0.060; AUC 0.93 vs 0.90

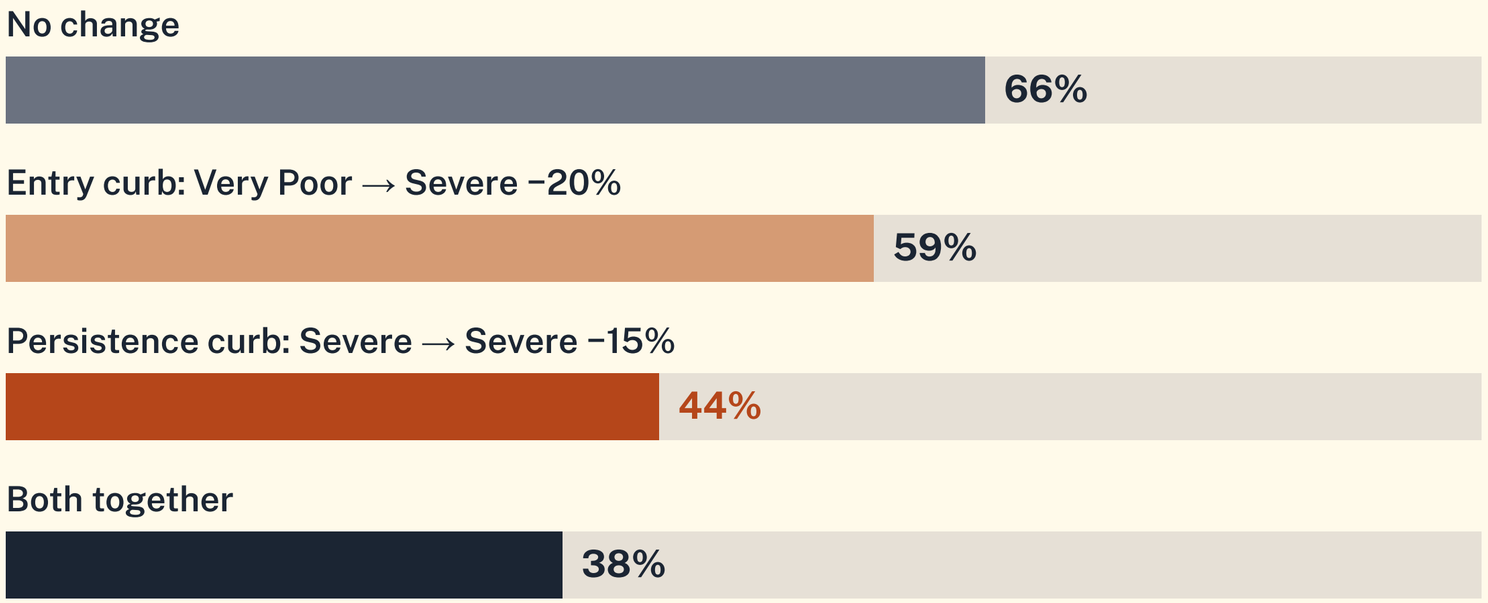
2+ days: Poisson has the lower Brier score; from 5 days its AUC is higher too

Why: $P^h \to \pi$, so the chain forgets today within a week, as on Vivian's slide

WHAT-IF SIMULATION · ASSUMED EFFECTS, NOT MEASURED ONES

Ending spells sooner beats delaying them

Winters with at least one 5-day Severe crisis (3,000 simulated 90-day winters each)



Severe days per winter

No change: **18.4**

Entry curb: **15.5**

Persistence curb: **15.4**

Both: **12.9**

Illustration only, using the winter matrix

CONCLUSION

The Poisson process says how many Severe days a winter will bring. The Markov chain says which days, and how long they last.

≈ 32 a year, 94% in Nov–Jan, no trend in 2016–2025

P(stay Severe) ≈ 0.63, both models give the same value

Tested on 2023–26: each model wins the question it was built for

INDIVIDUAL CONTRIBUTIONS

Who did what

DATA & POISSON

Harsh Raj

- CPCB download and cleaning: 18 hourly files into one daily dataset
- Exploratory analysis and the Severe threshold
- Homogeneous, seasonal and clustered Poisson models
- Likelihood-ratio, time-rescaling and out-of-sample tests

notebooks 01–02

MARKOV CHAIN

Vivian Mathew

- Five-state chain with seasonal transition matrices
- Homogeneity and order tests
- Stationary distribution, spell lengths, hitting times
- Season-start model and day-ahead validation

notebook 03

SIMULATION & VALIDATION

Rudraksh Kapil

- One simulation engine for both models
- Parameter-recovery check
- 2,000-decade scoreboard and extreme spells
- Multi-day forecast test and policy what-ifs

notebook 04

github.com/HarshRaj797864/asm-air-pollution · 4 notebooks · run top to bottom · fixed seed

Questions

Data: CPCB CAAQMS repository, R K Puram (DPCC site 124), 2016 – Sep 2026

Harsh Raj · Vivian Mathew · Rudraksh Kapil